follow a single narrow (mitochondrial) thread, as opposed to the ever branching thread of 'whole organism pedigrees'. Do the same for the Queen, following her narrow maternal thread back through the generations. Sooner or later the two threads will meet and now, by simply counting generations along the two threads, you can easily calculate your mitochondrial cousinship to the Queen.

What you can do for mitochondria, you can in principle do for any particular gene, and this illustrates the difference between a gene's point of view and an organism's point of view. From a whole organism point of view you have two parents, four grandparents, eight great grandparents, etc. But, like a mitochondrion, each gene has only one parent, one grandparent, one great grandparent, etc. I have one gene for blue eyes and the Queen has two. In principle we could trace the generations back and discover the cousinship between my blue eye gene and each of the Queen's two. The common ancestor of our two genes is called the 'coalescence point'. Coalescence analysis has become a flourishing branch of genetics and very fascinating it is. Can you see how congenial it is to the 'gene's eye view' that this whole book espouses? We are not talking about altruism any more. The gene's eye view is flexing its muscles in other domains, in this case looking back at ancestry.

You can even investigate the coalescence point between two alleles in one individual body. Prince Charles has blue eyes and we can assume that he has a pair of blue eye alleles opposite each other on Chromosome 15. How closely related to each other are Prince Charles' two blue eye genes, one from his father, one from his mother? In this case we know one possible answer, only because royal pedigrees are documented in ways that most of our pedigrees are not. Queen Victoria had blue eyes and Prince Charles is descended from Victoria in two ways: via King Edward VII on his mother's side; and via Princess Alice of Hesse on his father's side. There's a 50 percent probability that one of Victoria's blue eye genes peeled off two copies of itself, one of which went to her son, Edward VII, and the other to her daughter Princess Alice. Further copies of these two sibling genes could easily have passed down the generations to Queen Elizabeth II on one side and Prince Philip on the other, whence they were reunited in Prince Charles. This would mean the 'coalescence' point of Charles's two genes was Victoria. We do not—cannot—know whether this actually is true for Charles' blue eye genes. But statistically